ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

PROJECT PLAN

Work Breakdown Structure (WBS), Activity Network Diagram, Critical Path Method & Milestone Schedule

SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

SECTION 1 & 2

1. Work Breakdown Structure (WBS) & 2. Network Diagram

The project plan sequences the 87 person-day development effort across 13 hierarchical work packages:

- 1.0 Project Initiation & Charter
- 2.0 Requirements Analysis
- 3.0 Business Requirements (BRD)
- 4.0 Software Requirements (SRS)
- 5.0 System Analysis & Boundary
- 6.0 UI/UX Design & Prototype
- 7.0 UML Design Suite
- 8.0 Architecture Design
- 9.0 Development Planning
- 10.0 Testing & SQA Planning
- 11.0 Deployment Planning
- 12.0 Master Documentation
- 13.0 Project Closure & Audit

SECTION 3

3. Critical Path Method (CPM) & Float Analysis

TABLE 3.1: CPM FORWARD & BACKWARD PASS SCHEDULE TABLE

ID	ACTIVITY DESCRIPTION	DURATION	PREDECESSORS	ES	EF	LS	LF	FLOAT	CRITICAL?
A	Requirements Analysis	5d	--	0	5	0	5	0	CRITICAL
B	BRD Formulation	4d	A	5	9	5	9	0	CRITICAL
C	SRS Formulation	6d	B	9	15	9	15	0	CRITICAL
D	System Architecture	5d	C	15	20	15	20	0	CRITICAL
E	UML Design Suite	6d	D	20	26	20	26	0	CRITICAL
F	UI/UX Design	8d	C	15	23	18	26	3	TF=3
G	Pet Profile Planning	14d	E, F	26	40	30	44	4	TF=4
H	Appointment Booking Planning	18d	E, F	26	44	26	44	0	CRITICAL
I	Vaccination Reminder Planning	9d	G	40	49	51	60	11	TF=11
J	Payment Workflow Planning	10d	H	44	54	50	60	6	TF=6
K	Clinic Dashboard Planning	16d	G, H	44	60	44	60	0	CRITICAL
L	QA & Test Planning	14d	I, J, K	60	74	60	74	0	CRITICAL
M	Documentation Review	5d	L	74	79	74	79	0	CRITICAL
N	Deployment & Training Planning	6d	M	79	85	79	85	0	CRITICAL

> **Critical Path:** A → B → C → D → E → H → K → L → M → N = **85 Working Days**. All durations represent planning assumptions for academic estimation.